Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/26/2026
PI Name:	Zhen, Shuyang Ph.D.		
Project Title:	Environmental Optimization and Cultivar Selection to Enhance Dwarf Tomato Growth and Stress Tolerance for Space Agriculture		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 Space Biology NNH24ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	04/01/2026	**End Date:**	03/31/2030
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA HQ
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Key Personnel Changes/Previous PI:			
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Task Description:	Fresh food production in space is essential for long-duration human missions. However, spaceflight environments pose unique challenges to crop production. Astronauts on the International Space Station rely primarily on prepackaged food, which decreases in nutritional value and has low culinary appeal. Fresh food remains a major constraint for future long-term space missions. Advancing space crop production will enable more sustainable space exploration, reduce reliance on resupply from Earth, and boost crew nutrition and morale. This ground-based Early Career Research Investigation aims to enhance the growth, nutritional quality, and stress tolerance of dwarf tomato, a nutrient-dense food crop, in spaceflight systems through cultivar selection and environmental optimization. Dwarf tomato cultivars will serve as model crops to gain mechanistic insight into how spaceflight stressors, particularly altered gravity and super-elevated CO2 levels, affect plant growth, physiology, yield, and fruit nutritional value. We further aim to investigate how genetic variation affects cultivar responses to these stressors. Our specific objectives are to: (1) identify dwarf tomato cultivars with suitable horticultural traits under spaceflight-relevant conditions, (2) characterize drought and flooding stress responses to select cultivars with improved water stress tolerance under altered gravity, (3) down-select cultivars with enhanced tolerance to combined stressors, including super-elevated CO2, drought, and flooding, and (4) optimize key environmental conditions, particularly light intensity and spectral quality, to further improve growth, yield, nutritional quality, and stress tolerance. Over 30 commercially available dwarf tomato cultivars will be initially screened in a climate-controlled growth facility to evaluate key horticultural traits, including crop size, vigor, crop cycle, fruit yield and nutritional content, harvest index, and susceptibility to physiological disorders (e.g., intumescence) under spaceflight-relevant environmental conditions. Cultivars with acceptable horticultural traits and no major stress symptoms will be further evaluated for drought and flooding tolerance using different substrate types and cultivation methods. Growth, survival, and physiological responses, including photosynthetic activities, will be assessed under stress conditions. To evaluate tolerance to elevated CO2, six down-selected cultivars with improved drought and flooding tolerance will be grown under three CO2 levels: 450, 1000, and 3000 μmol mol-1. Growth and physiological parameters, including assimilation (A)-leaf internal CO2 (Ci) response curves (A/Ci curves) and leaf carbohydrate concentrations, will be examined to gain insight into the underlying stress-response mechanisms. In the final phase, key environmental factors, especially light intensity and spectral quality, will be optimized to further improve the growth, yield, fruit quality, and stress tolerance of top-performing cultivars. Potential impact: This project will generate foundational data on growth, yield, and nutritional quality of approximately 30 dwarf tomato cultivars and identify cultivars with improved tolerance to key spaceflight environmental stressors, including drought, flooding, and super-elevated CO2. It will provide mechanistic insights into how genetic variation influences stress responses and how environmental factors such as substrate type, cultivation method, light intensity, and spectral quality can be optimized to improve plant growth, stress tolerance, yield, and nutritional quality. Outcomes will enhance the production of fresh, nutrient-rich food to support crew health during long-duration space missions. Additionally, the results will have broader impacts by advancing resource-efficient controlled environment agriculture on Earth, thereby strengthening sustainable food production and enhancing food and nutrition security.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will generate foundational data on growth, yield, and nutritional quality of approximately 30 dwarf tomato cultivars and identify cultivars with improved tolerance to key spaceflight environmental stressors, including drought, flooding/hypoxia, and super-elevated CO2. It will provide mechanistic insights into how genetic variation influences stress responses and how environmental factors such as substrate type, cultivation method, light intensity, and spectral quality can be optimized to improve plant growth, stress tolerance, yield, and nutritional quality. Outcomes will enhance the production of fresh, nutrient-rich food to support crew health during long-duration space missions. The results will have broader impacts by advancing resource-efficient controlled environment agriculture on Earth, thereby strengthening sustainable food production and enhancing food and nutrition security.
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated:)